ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2025 P1H Worked Solutions

Nov 2025 | P1H

37 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 4

TOPIC: **Statistics Toolkit**

Nov 2025 P1H - Nov 2025 | P1H

1 The table gives information about the number of kilometres that Ted cycled on each of the 30 days in April.

Number of kilometres (K)	**Frequency**
$0 \leqslant K < 5$	8
$5 \leqslant K < 10$	7
$10 \leqslant K < 15$	3
$15 \leqslant K < 20$	10
$20 \leqslant K < 25$	2

Calculate an estimate for the mean number of kilometres that Ted cycled on each day in April.

.... kilometres

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Statistics Toolkit**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Statistics Toolkit. The tag is correct.

Solution

Use midpoints $2.5, 7.5, 12.5, 17.5, 22.5$.

$$2.5(8) + 7.5(7) + 12.5(3) + 17.5(10) + 22.5(2) = 330$$

$$\text{estimated mean} = \frac{330}{30} = 11$$

FINAL ANSWER

11 km.

PAST PAPER QUESTION Q#: 2 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Nov 2025 P1H - Nov 2025 | P1H

2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

(Total for Question 2 is 2 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

Round each number to one significant figure:

$$2.11 \approx 2, \quad 58.9 \approx 60, \quad 8.859 \approx 9$$

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}} \approx \frac{2^2 \times 60}{\sqrt{9}}$$
$$= \frac{4 \times 60}{3} = 80$$

FINAL ANSWER

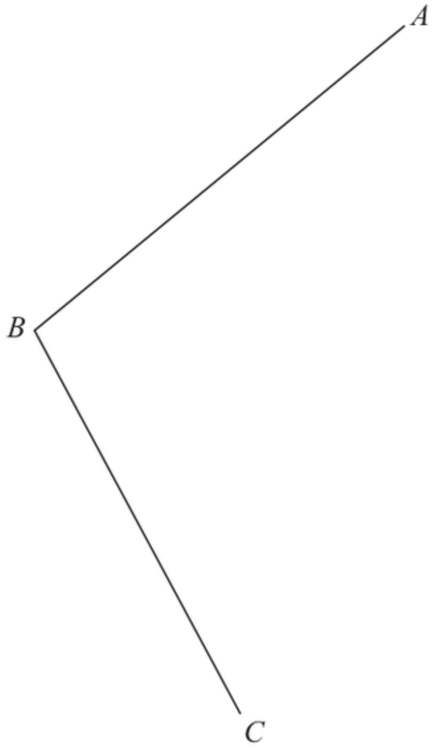
80

PAST PAPER QUESTION Q#: 3 Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

Nov 2025 P1H - Nov 2025 | P1H

3 Using ruler and compasses only, construct the bisector of angle ABC
Show all your construction lines.



(Total for Question 3 is 2 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Bearings, Scale Drawing & Constructions. The tag is correct.

Construction answer

To construct the bisector of angle ABC:

1. Use centre B to draw an arc crossing BA and BC.
2. Use the same compass width from the two crossing points to make two arcs inside the angle.
3. Join B to the point where these two arcs meet.

FINAL ANSWER

That line is the required angle bisector.

PAST PAPER QUESTION Q#: 4 Marks: 3

TOPIC: **Statistics Toolkit**

Nov 2025 P1H - Nov 2025 | P1H

4 e, f and g represent integers such that

$$e \qquad f \qquad 7 \qquad g \qquad 12 \qquad 17$$

is a list of integers written in order of size.

The integers have

a range of 15
a median of 8.5
a mean of 9

Work out the value of e, the value of f and the value of g

e = ...

f = ...

g = ...

(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 3

TOPIC: **Statistics Toolkit**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The range is 15:

$$17 - e = 15$$

$$e = 2$$

The mean is 9, so the total is

$$6 \times 9 = 54$$

$$2 + f + 7 + g + 12 + 17 = 54$$

$$f + g = 16$$

The median is 8.5:

$$\frac{7 + g}{2} = 8.5$$

$$g = 10$$

$$f = 16 - 10 = 6$$

FINAL ANSWER

$e = 2,\ f = 6,\ g = 10.$

PAST PAPER QUESTION

Q#: 5 Marks: 3

TOPIC: **Algebraic Proof**

Nov 2025 P1H - Nov 2025 | P1H

5 Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 3

TOPIC: **Algebraic Proof**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic proof. The tag is acceptable because the question asks to show the exact result.

Solution

$$\frac{3}{7}+\frac{1}{3}$$

Use denominator 21:

$$\frac{3}{7}=\frac{9}{21}, \qquad \frac{1}{3}=\frac{7}{21}$$

$$\frac{9}{21}+\frac{7}{21}=\frac{16}{21}$$

FINAL ANSWER

$\frac{16}{21}$.

PAST PAPER QUESTION Q#: 6 Marks: 4

TOPIC: **Ratio Problem Solving**

Nov 2025 P1H - Nov 2025 | P1H

6 Eli and Peta share \$275 in the ratio 2 : 3

Eli gives $\frac{3}{11}$ of his share to charity.

Peta gives 0.32 of her share to charity.

Work out the total amount that Eli and Peta give to charity.

\$...

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 4

TOPIC: **Ratio Problem Solving**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Ratio Problem Solving. The tag is correct.

Method

The ratio is

$$\text{Eli} : \text{Peta} = 2 : 3$$

There are 5 parts, so one part is

$$275 \div 5 = 55$$

Eli gets

$$2 \times 55 = 110$$

Peta gets

$$3 \times 55 = 165$$

Eli gives

$$\frac{3}{11} \times 110 = 30$$

Peta gives

$$0.32 \times 165 = 52.8$$

Total given to charity:

$$30 + 52.8 = 82.8$$

FINAL ANSWER

82.8.

PAST PAPER QUESTION

Q#: 7 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Nov 2025 P1H - Nov 2025 | P1H

7 The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

.... g

(1)

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

.... cm

(1)

(Total for Question 7 is 2 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

The weight 475 g correct to the nearest g has lower bound

$$475 - 0.5 = 474.5$$

The height 120 cm correct to the nearest 10 cm has upper bound

$$120 + 5 = 125$$

FINAL ANSWER

(a) 474.5 g, (b) 125 cm

PAST PAPER QUESTION

Q#: 8 Marks: 2

TOPIC: **Algebraic Roots & Indices**

Nov 2025 P1H - Nov 2025 | P1H

8 $\dfrac{8^{-2} \times 8^{9}}{8^{10}} = 8^{n}$

Work out the value of n

$n =$ ....

(Total for Question 8 is 2 marks)

PAST PAPER SOLUTION Q#: 8 Marks: 2

TOPIC: **Algebraic Roots & Indices**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Algebraic roots and indices. The tag is correct.

Solution

$$\frac{8^{-2} \times 8^{9}}{8^{10}} = 8^{-2+9-10} = 8^{-3}$$

FINAL ANSWER

$n = -3$

PAST PAPER QUESTION Q#: 9 Marks: 3

Topic: **Percentages**

Nov 2025 P1H - Nov 2025 | P1H

9 Hari scored 140 points playing a game on Tuesday.
This was 12% more points than he scored playing the game on Monday.

Work out the number of points that Hari scored playing the game on Monday.

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Percentages**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Percentages. The tag is correct.

Method

Tuesday's score was 12% more than Monday's score, so Tuesday was 112% of Monday.

$$1.12 \times \text{Monday} = 140$$

$$\text{Monday} = \frac{140}{1.12} = 125$$

FINAL ANSWER

125 points

PAST PAPER QUESTION Q#: 10 Marks: 7

Topic: **Factorising**

Nov 2025 P1H - Nov 2025 | P1H

10 (a) Simplify $(6m)^0$

....

(1)

(b) Solve $5y + 20 < 7y + 1$

....

(2)

(c) Factorise fully $15w^2x^5 + 25w^3x^2$

....

(2)

(d) Simplify fully $\dfrac{36a^5c^7}{12a^2c^3}$

....

(2)

(Total for Question 10 is 7 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 7

TOPIC: **Factorising**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Factorising. The tag is acceptable for this mixed algebra question.

Solution

(a) $(6m)^0 = 1$.

(b)

$$5y + 20 < 7y + 1$$

$$19 < 2y$$

$$y > \frac{19}{2}$$

(c)

$$15w^2x^5 + 25w^3x^2 = 5w^2x^2(3x^3 + 5w)$$

(d)

$$\frac{36a^5c^7}{12a^2c^3} = 3a^3c^4$$

FINAL ANSWER

1, $y > \frac{19}{2}$, $5w^2x^2(3x^3 + 5w)$, $3a^3c^4$

PAST PAPER QUESTION Q#: 11 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2025 P1H - Nov 2025 | P1H

11 *ABC* is a right-angled triangle.

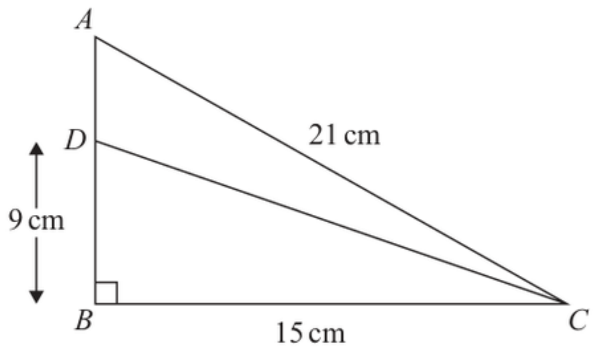

Diagram **NOT** accurately drawn

$AC = 21$ cm $\quad BC = 15$ cm $\quad$ angle $ABC = 90°$

The point D lies on AB such that $DB = 9$ cm

Work out the size of angle ACD
Give your answer correct to one decimal place.

....°

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11 **Marks: 4**

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.

Solution

First find AB:

$$AB^2 = 21^2 - 15^2 = 216$$

$$AB = 14.696\ldots$$

$$AD = AB - DB = 14.696\ldots - 9 = 5.696\ldots$$

Angle ACB:

$$\cos ACB = \frac{15}{21}$$

$$ACB = 44.415\ldots^\circ$$

Angle DCB:

$$\tan DCB = \frac{9}{15}$$

$$DCB = 30.963\ldots^\circ$$

$$ACD = 44.415\ldots - 30.963\ldots = 13.451\ldots^\circ$$

FINAL ANSWER

13.5°.

PAST PAPER QUESTION

Q#: 12 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Nov 2025 P1H - Nov 2025 | P1H

12

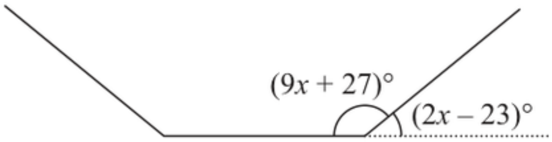

Diagram **NOT** accurately drawn

The diagram shows part of a regular n-sided polygon with

an interior angle of $(9x + 27)°$
an exterior angle of $(2x – 23)°$

Work out the value of n

n = ...

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Angles in Polygons & Parallel Lines. The tag is correct.

Solution

The interior angle and exterior angle of a regular polygon add to 180°:

$$(9x + 27) + (2x - 23) = 180$$

$$11x + 4 = 180$$

$$x = 16$$

The exterior angle is

$$2x - 23 = 2(16) - 23 = 9^\circ$$

For a regular polygon,

$$n = \frac{360}{9} = 40$$

FINAL ANSWER

$n = 40$.

PAST PAPER QUESTION

Q#: 13 Marks: 5

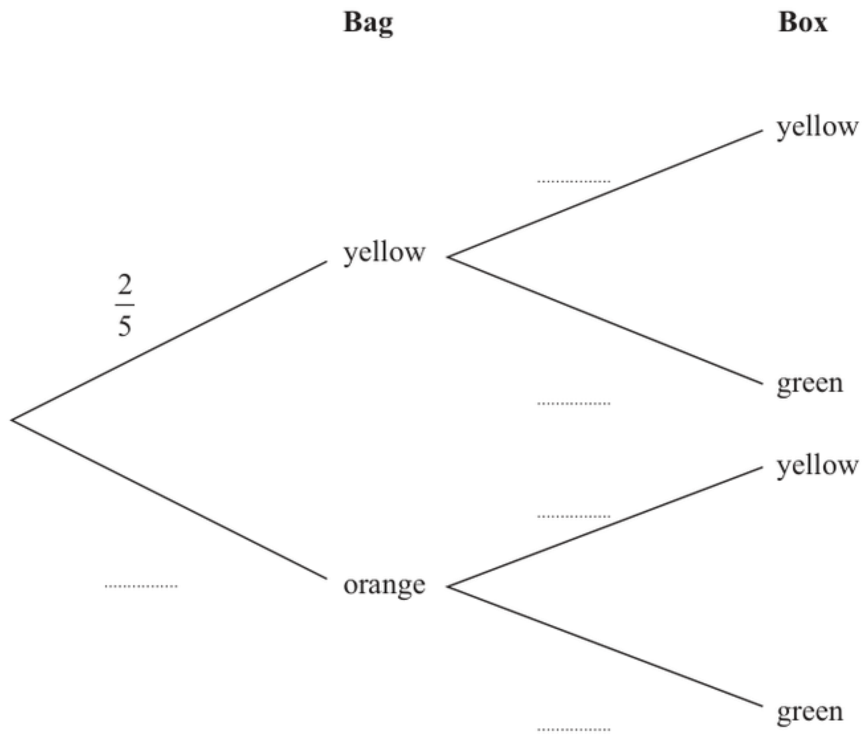
TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Nov 2025 P1H - Nov 2025 | P1H

13 In a bag, there are only 2 yellow beads and 3 orange beads.
In a box, there are only 3 yellow beads and 5 green beads.

Chi takes at random a bead from the bag and a bead from the box.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Chi takes two beads of different colours.

....

(3)

(Total for Question 13 is 5 marks)

PAST PAPER SOLUTION Q#: 13 Marks: 5

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

Bag 1:

$$P(Y) = \frac{2}{5}, \qquad P(O) = \frac{3}{5}$$

Bag 2:

$$P(Y) = \frac{3}{8}, \qquad P(G) = \frac{5}{8}$$

The only way to get the same colour is yellow then yellow.

$$P(\text{different}) = 1 - \frac{2}{5} \times \frac{3}{8}$$

$$= 1 - \frac{3}{20} = \frac{17}{20}$$

FINAL ANSWER

$\frac{17}{20}$.

PAST PAPER QUESTION

TOPIC: **Expanding Brackets**

Nov 2025 P1H - Nov 2025 | P1H

Q#: 14 Marks: 6

14 (a) Expand and simplify $7x(3x + 2)(2x - 5)$

....

(3)

(b) Solve $\dfrac{9}{2y} + \dfrac{5}{7} = 5$

Show clear algebraic working.

$y =$ ....

(3)

(Total for Question 14 is 6 marks)

PAST PAPER SOLUTION

Q#: 14 **Marks: 6**

TOPIC: **Expanding Brackets**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Expanding brackets. The tag is acceptable.

Solution

(a)

$$7x(3x+2)(2x-5) = 7x(6x^2 - 11x - 10)$$

$$= 42x^3 - 77x^2 - 70x$$

(b)

$$\frac{9}{2y} + \frac{5}{7} = 5$$

$$\frac{9}{2y} = \frac{30}{7}$$

$$63 = 60y$$

FINAL ANSWER

(a) $42x^3 - 77x^2 - 70x$, (b) $y = \frac{21}{20}$

PAST PAPER QUESTION

TOPIC: **Circles, Arcs & Sectors**

Nov 2025 P1H - Nov 2025 | P1H

Q#: 15 Marks: 2

15 *AOB* is a sector of a circle with centre *O* and radius 12 cm

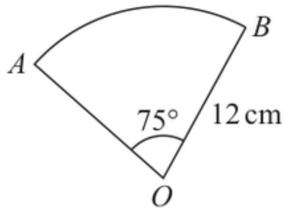

Diagram **NOT** accurately drawn

Work out the area of the sector.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 15 is 2 marks)

PAST PAPER SOLUTION Q#: 15 Marks: 2

TOPIC: **Circles, Arcs & Sectors**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Circles, Arcs & Sectors. The tag is correct.

Solution

$$\text{area} = \frac{75}{360} \times \pi(12)^2$$

$$= 94.247\ldots$$

FINAL ANSWER

$94.2\ \text{cm}^2$.

PAST PAPER QUESTION Q#: 16 Marks: 2

TOPIC: **Statistics Toolkit**

Nov 2025 P1H - Nov 2025 | P1H

16 Ivan asked 15 people how many books they read last year.

Here are his results.

1 1 3 4 6 7 8 9 10 12 15 25 30 37 50

Work out the interquartile range of the number of books read.

(Total for Question 16 is 2 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 2

TOPIC: **Statistics Toolkit**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The data are already in order:

$$1,\ 1,\ 3,\ 4,\ 6,\ 7,\ 8,\ 9,\ 10,\ 12,\ 15,\ 25,\ 30,\ 37,\ 50$$

There are 15 values.

$$Q_1 = 4, \qquad Q_3 = 25$$

$$\text{IQR} = 25 - 4 = 21$$

FINAL ANSWER

21.

PAST PAPER QUESTION

Q#: 17 **Marks: 3**

TOPIC: **Algebraic Proof**

Nov 2025 P1H - Nov 2025 | P1H

17 Show that $\dfrac{2\sqrt{7}+2}{\sqrt{7}-3}$ can be written in the form $a-\sqrt{b}$ where a and b are integers.

Show your working clearly.

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Algebraic Proof**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Algebraic proof. The tag is acceptable because the question asks for exact-form proof.

Solution

$$\frac{2\sqrt{7}+2}{\sqrt{7}-3} \times \frac{\sqrt{7}+3}{\sqrt{7}+3} = \frac{(2\sqrt{7}+2)(\sqrt{7}+3)}{7-9}$$

$$= \frac{14+6\sqrt{7}+2\sqrt{7}+6}{-2}$$

$$= \frac{20+8\sqrt{7}}{-2} = -10-4\sqrt{7}$$

Since $4\sqrt{7} = \sqrt{112}$,

$$-10-4\sqrt{7} = -10-\sqrt{112}$$

FINAL ANSWER

$-10-\sqrt{112}$, so $a = -10$ and $b = 112$.

PAST PAPER QUESTION

Q#: 18 **Marks: 4**

TOPIC: **Rearranging Formulas**

Nov 2025 P1H - Nov 2025 | P1H

18 Make y the subject of $p = \sqrt{\dfrac{y + w}{3y - t}}$

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: **Rearranging Formulas**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Rearranging formulas. The tag is correct.

Solution

$$p = \sqrt{\frac{y+w}{3y-t}}$$

$$p^2 = \frac{y+w}{3y-t}$$

$$p^2(3y-t) = y+w$$

$$3p^2y - p^2t = y+w$$

$$y(3p^2-1) = w + p^2t$$

FINAL ANSWER

$y = \frac{w+p^2t}{3p^2-1}$

PAST PAPER QUESTION Q#: 19 Marks: 3

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Nov 2025 P1H - Nov 2025 | P1H

19 The diagram shows triangle *ABC*

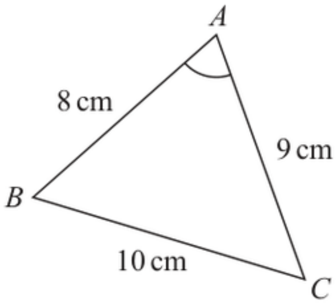

Diagram **NOT** accurately drawn

$AB = 8$ cm $\quad BC = 10$ cm $\quad CA = 9$ cm

Work out the size of angle *BAC*
Give your answer correct to one decimal place.

.... °

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 **Marks: 3**

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Corrected to Sine, Cosine Rule & Area of Triangles.

Solution

Use the cosine rule for angle BAC:

$$\cos A = \frac{8^2 + 9^2 - 10^2}{2(8)(9)}$$

$$\cos A = \frac{45}{144}$$

$$A = 71.790\ldots^\circ$$

FINAL ANSWER

71.8°.

PAST PAPER QUESTION

Q#: 20 Marks: 3

TOPIC: **Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**
Give your answer in the form $y = mx + c$

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION Q#: 20 Marks: 3

TOPIC: **Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Correct.

Method

Line L has equation

$$y = 4x + 7$$

So its gradient is

$$4$$

Line M is perpendicular to L, so its gradient is

$$-\frac{1}{4}$$

Line M passes through $(8, 1)$.

$$y - 1 = -\frac{1}{4}(x - 8)$$

$$y - 1 = -\frac{1}{4}x + 2$$

$$y = -\frac{1}{4}x + 3$$

FINAL ANSWER

$y = -\frac{1}{4}x + 3$

PAST PAPER QUESTION Q#: 20 Marks: 3

TOPIC: **Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**
Give your answer in the form $y = mx + c$

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 **Marks: 3**

TOPIC: **Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Correct.

Method

Line L has equation

$$y = 4x + 7$$

So its gradient is

$$4$$

Line M is perpendicular to L, so its gradient is

$$-\frac{1}{4}$$

Line M passes through $(8, 1)$.

$$y - 1 = -\frac{1}{4}(x - 8)$$

$$y - 1 = -\frac{1}{4}x + 2$$

$$y = -\frac{1}{4}x + 3$$

FINAL ANSWER

$y = -\frac{1}{4}x + 3$

PAST PAPER QUESTION

Q#: 21 **Marks: 3**

TOPIC: **Completing the Square**

Nov 2025 P1H - Nov 2025 | P1H

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a, b and c are integers.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Completing the Square**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Correct.

Method

$$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$$

$$= 5\left((x-2)^2 - 4\right) + 23$$

$$= 5(x-2)^2 - 20 + 23$$

$$= 5(x-2)^2 + 3$$

FINAL ANSWER

$5(x-2)^2 + 3$

PAST PAPER QUESTION

Q#: 21 Marks: 3

TOPIC: **Completing the Square**

Nov 2025 P1H - Nov 2025 | P1H

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a, b and c are integers.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Completing the Square**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$$

$$= 5\left((x-2)^2 - 4\right) + 23$$

$$= 5(x-2)^2 - 20 + 23$$

$$= 5(x-2)^2 + 3$$

FINAL ANSWER

$5(x-2)^2 + 3$

PAST PAPER QUESTION Q#: 22 Marks: 3

TOPIC: **Combined & Conditional Probability**

Nov 2025 P1H - Nov 2025 | P1H

22 There are 15 buttons in a box.

3 of the buttons are red
2 of the buttons are pink
10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 22 is 3 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 3

TOPIC: **Combined & Conditional Probability**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

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Topic check. Correct.

Method

There are 2 pink buttons.
At least one pink button is still in the box unless both pink buttons are taken.
So use the complement.
The probability that both pink buttons are taken is

$$\frac{\binom{2}{2}\binom{13}{1}}{\binom{15}{3}}$$

$$= \frac{13}{455}$$

$$= \frac{1}{35}$$

Therefore,

$$P(\text{at least one pink is still in the box}) = 1 - \frac{1}{35}$$

$$= \frac{34}{35}$$

FINAL ANSWER

$\frac{34}{35}$

PAST PAPER QUESTION

Q#: 22 Marks: 3

TOPIC: **Combined & Conditional Probability**

Nov 2025 P1H - Nov 2025 | P1H

22 There are 15 buttons in a box.

3 of the buttons are red
2 of the buttons are pink
10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 22 is 3 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 3

TOPIC: **Combined & Conditional Probability**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

There are 2 pink buttons.
At least one pink button is still in the box unless both pink buttons are taken.
So use the complement.
The probability that both pink buttons are taken is

$$\frac{\binom{2}{2}\binom{13}{1}}{\binom{15}{3}}$$

$$= \frac{13}{455}$$

$$= \frac{1}{35}$$

Therefore,

$$P(\text{at least one pink is still in the box}) = 1 - \frac{1}{35}$$

$$= \frac{34}{35}$$

FINAL ANSWER

$\frac{34}{35}$

PAST PAPER QUESTION Q#: 23 Marks: 3

TOPIC: **Algebraic Fractions**

Nov 2025 P1H - Nov 2025 | P1H

23 $a = \dfrac{2x+5}{1-x}$ $\quad x = \dfrac{5-2y}{3y}$

Write a in the form $\dfrac{m+ny}{p(y-1)}$ where m, n and p are integers.

Show your working clearly.

$a =$ ....

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 3

TOPIC: **Algebraic Fractions**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Ratio Toolkit to Algebraic Fractions.

Method

$$a = \frac{2x+5}{1-x}, \quad x = \frac{5-2y}{3y}$$

Substitute for x.
Numerator:

$$2x + 5 = 2\left(\frac{5-2y}{3y}\right) + 5$$

$$= \frac{10 - 4y + 15y}{3y} = \frac{10 + 11y}{3y}$$

Denominator:

$$1 - x = 1 - \frac{5-2y}{3y}$$

$$= \frac{3y - 5 + 2y}{3y} = \frac{5y-5}{3y} = \frac{5(y-1)}{3y}$$

Therefore

$$a = \frac{\frac{10+11y}{3y}}{\frac{5(y-1)}{3y}} = \frac{10+11y}{5(y-1)}$$

FINAL ANSWER

$$\frac{10+11y}{5(y-1)}$$

PAST PAPER QUESTION Q#: 23 Marks: 3

TOPIC: **Algebraic Fractions**

Nov 2025 P1H - Nov 2025 | P1H

23 $a = \dfrac{2x+5}{1-x} \qquad x = \dfrac{5-2y}{3y}$

Write a in the form $\dfrac{m+ny}{p(y-1)}$ where m, n and p are integers.

Show your working clearly.

$a =$ ....

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **Algebraic Fractions**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Ratio Toolkit to Algebraic Fractions.

Method

$$a = \frac{2x+5}{1-x}, \quad x = \frac{5-2y}{3y}$$

Substitute for x.
Numerator:

$$2x+5 = 2\left(\frac{5-2y}{3y}\right)+5$$

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Denominator:

$$1-x = 1-\frac{5-2y}{3y}$$

$$= \frac{3y-5+2y}{3y} = \frac{5y-5}{3y} = \frac{5(y-1)}{3y}$$

Therefore

$$a = \frac{\frac{10+11y}{3y}}{\frac{5(y-1)}{3y}} = \frac{10+11y}{5(y-1)}$$

FINAL ANSWER

$$\frac{10+11y}{5(y-1)}$$

PAST PAPER QUESTION Q#: 24 Marks: 3

TOPIC: **Transformations of Graphs**

Nov 2025 P1H - Nov 2025 | P1H

24 A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x + 5)$

(.............................. , ...)
(1)

(ii) $y = f(x) + 7$

(.............................. , ...)
(1)

(iii) $y = f(2x)$

(.............................. , ...)
(1)

(Total for Question 24 is 3 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 3

TOPIC: **Transformations of Graphs**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The turning point of $y = f(x)$ is $(4, 3)$.
For

$$y = f(x + 5)$$

the graph moves 5 units left:

$$(4, 3) \to (-1, 3)$$

For

$$y = f(x) + 7$$

the graph moves 7 units up:

$$(4, 3) \to (4, 10)$$

For

$$y = f(2x)$$

the x-coordinate is divided by 2:

$$(4, 3) \to (2, 3)$$

FINAL ANSWER

$(-1, 3)$, $(4, 10)$, and $(2, 3)$

PAST PAPER QUESTION Q#: 24 Marks: 3

TOPIC: **Transformations of Graphs**

Nov 2025 P1H - Nov 2025 | P1H

24 A curve has equation $y = \mathrm{f}(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i) $y = \mathrm{f}(x + 5)$

(.............................. , ...)
(1)

(ii) $y = \mathrm{f}(x) + 7$

(.............................. , ...)
(1)

(iii) $y = \mathrm{f}(2x)$

(.............................. , ...)
(1)

(Total for Question 24 is 3 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 3

TOPIC: **Transformations of Graphs**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The turning point of $y = f(x)$ is $(4, 3)$.
For

$$y = f(x + 5)$$

the graph moves 5 units left:

$$(4, 3) \to (-1, 3)$$

For

$$y = f(x) + 7$$

the graph moves 7 units up:

$$(4, 3) \to (4, 10)$$

For

$$y = f(2x)$$

the x-coordinate is divided by 2:

$$(4, 3) \to (2, 3)$$

FINAL ANSWER

$(-1, 3)$, $(4, 10)$, and $(2, 3)$

PAST PAPER QUESTION Q#: 25 Marks: 3

TOPIC: **Solving Inequalities**

Nov 2025 P1H - Nov 2025 | P1H

25 Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Solving Inequalities**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$2x^2 + x - 28 > 0$$

Factorise:

$$2x^2 + x - 28 = (2x - 7)(x + 4)$$

The critical values are

$$x = \frac{7}{2} \quad \text{and} \quad x = -4$$

The quadratic is greater than 0 outside the roots.

FINAL ANSWER

$x < -4$ or $x > \frac{7}{2}$

PAST PAPER QUESTION

Q#: 25 Marks: 3

TOPIC: **Solving Inequalities**

Nov 2025 P1H - Nov 2025 | P1H

25 Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Solving Inequalities**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$2x^2 + x - 28 > 0$$

Factorise:

$$2x^2 + x - 28 = (2x - 7)(x + 4)$$

The critical values are

$$x = \frac{7}{2} \quad \text{and} \quad x = -4$$

The quadratic is greater than 0 outside the roots.

FINAL ANSWER

$x < -4$ or $x > \frac{7}{2}$

PAST PAPER QUESTION Q#: 26 Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Nov 2025 P1H - Nov 2025 | P1H

26 Here is a prism *ABCDEFGH* with a horizontal, rectangular base *ABGF*

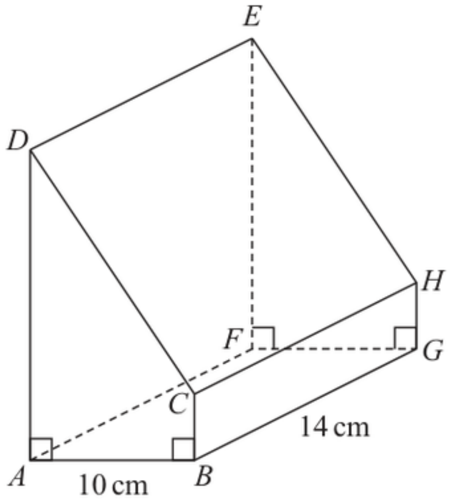


Diagram **NOT** accurately drawn

$AB = 10\,\text{cm}$ $BG = 14\,\text{cm}$

angle DAB = angle ABC = angle EFG = angle $FGH = 90°$

$BC = \dfrac{1}{5}AD$

The angle of elevation of E from B is $50°$

Calculate the volume of the prism.
Give your answer correct to 3 significant figures.
Show your working clearly.

.... cm^3

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26 **Marks: 5**

TOPIC: **3D Pythagoras & Trigonometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Right-Angled Triangles to 3D Pythagoras & Trigonometry.

Method

The horizontal base $ABGF$ is a rectangle.

$$AB = 10, \qquad BG = 14$$

So the horizontal distance from B to F is

$$BF = \sqrt{10^2 + 14^2} = \sqrt{296}$$

The angle of elevation of E from B is $50°$.
Since E is vertically above F,

$$\tan 50° = \frac{FE}{BF}$$

$$FE = \sqrt{296}\tan 50°$$

The prism has the same cross section throughout, so

$$AD = FE = \sqrt{296}\tan 50°$$

Also

$$BC = \frac{1}{5}AD$$

Area of the trapezium cross section $ABCD$:

$$\frac{1}{2}(AD + BC) \times AB$$

$$= \frac{1}{2}\left(AD + \frac{1}{5}AD\right) \times 10$$

$$= 6AD$$

Volume of the prism:

$$6AD \times 14 = 84AD$$

$$= 84\sqrt{296}\tan 50°$$

$$= 1722.311\ldots$$

Correct to 3 significant figures:

$$1720$$

FINAL ANSWER

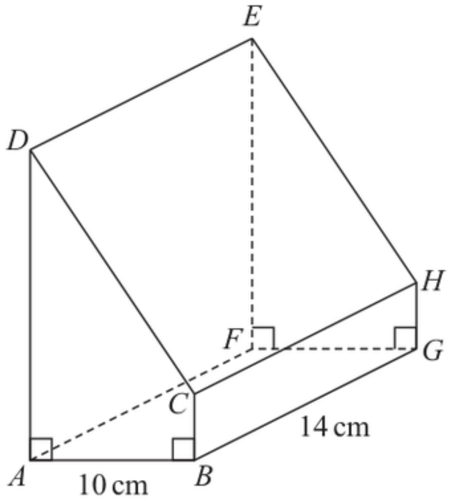
$\boxed{1720\ \text{cm}^3}$

PAST PAPER QUESTION Q#: 26 Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Nov 2025 P1H - Nov 2025 | P1H

26 Here is a prism *ABCDEFGH* with a horizontal, rectangular base *ABGF*



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angle DAB = angle ABC = angle EFG = angle $FGH = 90°$

$BC = \frac{1}{5}AD$

The angle of elevation of E from B is $50°$

Calculate the volume of the prism.
Give your answer correct to 3 significant figures.
Show your working clearly.

.... cm^3

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION Q#: 26 Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Right-Angled Triangles to 3D Pythagoras & Trigonometry.

Method

The horizontal base $ABGF$ is a rectangle.

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$$= 6AD$$

Volume of the prism:

$$6AD \times 14 = 84AD$$

$$= 84\sqrt{296}\tan 50°$$

$$= 1722.311\ldots$$

Correct to 3 significant figures:

$$1720$$

FINAL ANSWER

$\boxed{1720 \text{ cm}^3}$

PAST PAPER QUESTION

Q#: 27 **Marks: 6**

TOPIC: **Vectors**

Nov 2025 P1H - Nov 2025 | P1H

27

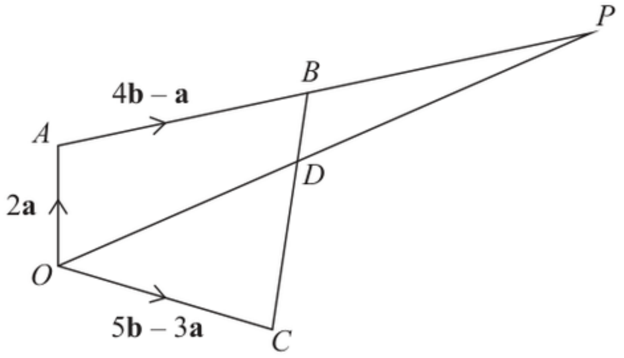


Diagram **NOT** accurately drawn

OABC is a quadrilateral.
ABP and *ODP* are straight lines.

$\overrightarrow{OA} = 2\mathbf{a}$ $\quad \overrightarrow{AB} = 4\mathbf{b} - \mathbf{a}$ $\quad \overrightarrow{OC} = 5\mathbf{b} - 3\mathbf{a}$

(a) Find an expression in terms of **a** and **b** for the vector $\overrightarrow{BC}$
Simplify your answer.

....
(2)

The point *D* lies on *BC* such that $BD : DC = 1 : 3$

Given that $\overrightarrow{OP} = n\overrightarrow{OD}$

(b) use a vector method to find the value of *n*

$n =$ ....
(4)

(Total for Question 27 is 6 marks)

PAST PAPER SOLUTION Q#: 27 Marks: 6

TOPIC: **Vectors**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Linear Graphs to Vectors.

Method

$$\vec{OA} = 2\mathbf{a}, \qquad \vec{AB} = 4\mathbf{b} - \mathbf{a}, \qquad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

First find $\vec{OB}$:

$$\vec{OB} = \vec{OA} + \vec{AB}$$

$$\vec{OB} = 2\mathbf{a} + 4\mathbf{b} - \mathbf{a}$$

$$\vec{OB} = \mathbf{a} + 4\mathbf{b}$$

For part (a),

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$\vec{BC} = (5\mathbf{b} - 3\mathbf{a}) - (\mathbf{a} + 4\mathbf{b})$$

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}$$

For part (b), $BD : DC = 1 : 3$, so

$$\vec{OD} = \vec{OB} + \frac{1}{4}\vec{BC}$$

$$\vec{OD} = \mathbf{a} + 4\mathbf{b} + \frac{1}{4}(\mathbf{b} - 4\mathbf{a})$$

$$\vec{OD} = \frac{17}{4}\mathbf{b}$$

Since A, B, P are collinear, let

$$\vec{OP} = \vec{OA} + s\vec{AB}$$

$$\vec{OP} = 2\mathbf{a} + s(4\mathbf{b} - \mathbf{a})$$

$$\vec{OP} = (2 - s)\mathbf{a} + 4s\mathbf{b}$$

But O, D, P are collinear, and $\vec{OD}$ has only a $\mathbf{b}$ component.
So

$$2 - s = 0$$

$$s = 2$$

Then

$$\vec{OP} = 8\mathbf{b}$$

Given

$$\vec{OP} = n\vec{OD}$$

$$8\mathbf{b} = n\left(\frac{17}{4}\mathbf{b}\right)$$

$$n = \frac{32}{17}$$

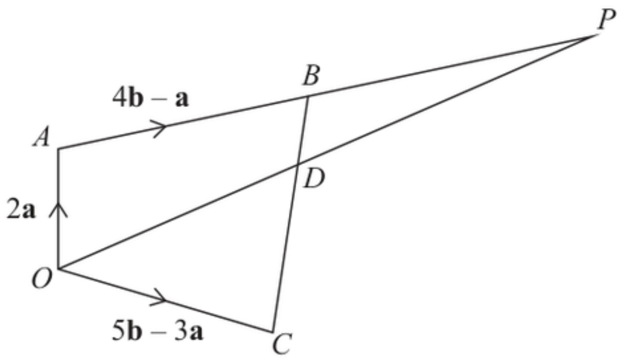
FINAL ANSWER
$\vec{BC} = \mathbf{b} - 4\mathbf{a}, \quad n = \dfrac{32}{17}$

PAST PAPER QUESTION Q#: 27 Marks: 6

TOPIC: **Vectors**

Nov 2025 P1H - Nov 2025 | P1H

27



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ABP and *ODP* are straight lines.

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Simplify your answer.

....
(2)

The point D lies on BC such that $BD : DC = 1 : 3$

Given that $\overrightarrow{OP} = n\overrightarrow{OD}$

(b) use a vector method to find the value of n

$n =$ ....
(4)

(Total for Question 27 is 6 marks)

PAST PAPER SOLUTION

Q#: 27 Marks: 6

TOPIC: **Vectors**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Linear Graphs to Vectors.

Method

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First find $\vec{OB}$:

$$\vec{OB} = \vec{OA} + \vec{AB}$$

$$\vec{OB} = 2\mathbf{a} + 4\mathbf{b} - \mathbf{a}$$

$$\vec{OB} = \mathbf{a} + 4\mathbf{b}$$

For part (a),

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$\vec{BC} = (5\mathbf{b} - 3\mathbf{a}) - (\mathbf{a} + 4\mathbf{b})$$

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}$$

For part (b), $BD : DC = 1 : 3$, so

$$\vec{OD} = \vec{OB} + \frac{1}{4}\vec{BC}$$

$$\vec{OD} = \mathbf{a} + 4\mathbf{b} + \frac{1}{4}(\mathbf{b} - 4\mathbf{a})$$

$$\vec{OD} = \frac{17}{4}\mathbf{b}$$

Since A, B, P are collinear, let

$$\vec{OP} = \vec{OA} + s\vec{AB}$$

$$\vec{OP} = 2\mathbf{a} + s(4\mathbf{b} - \mathbf{a})$$

$$\vec{OP} = (2 - s)\mathbf{a} + 4s\mathbf{b}$$

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Then

$$\vec{OP} = 8\mathbf{b}$$

Given

$$\vec{OP} = n\vec{OD}$$

$$8\mathbf{b} = n\left(\frac{17}{4}\mathbf{b}\right)$$

$$n = \frac{32}{17}$$

FINAL ANSWER

$\vec{BC} = \mathbf{b} - 4\mathbf{a}, \quad n = \dfrac{32}{17}$

PAST PAPER QUESTION Q#: 28 Marks: 6

TOPIC: **Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

28 The curve **C** has equation $x^2 + y^2 = d - 11x$ where d is an integer.
The line **L** has equation $y = 3x + e$ where e is an integer.

C and **L** intersect at the point A and at the point B

The coordinates of A are (0.2, 2.6)

Work out the coordinates of B
Show your working clearly.

(.......................... , ...)

(Total for Question 28 is 6 marks)

PAST PAPER SOLUTION Q#: 28 Marks: 6

TOPIC: **Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The line is

$$y = 3x + e$$

Point $A = (0.2, 2.6)$ lies on the line.

$$2.6 = 3(0.2) + e$$

$$2.6 = 0.6 + e$$

$$e = 2$$

So

$$y = 3x + 2$$

The curve is

$$x^2 + y^2 = d - 11x$$

Use point $A = (0.2, 2.6)$:

$$0.2^2 + 2.6^2 = d - 11(0.2)$$

$$0.04 + 6.76 = d - 2.2$$

$$d = 9$$

Now substitute $y = 3x + 2$ into the curve:

$$x^2 + (3x + 2)^2 = 9 - 11x$$

$$x^2 + 9x^2 + 12x + 4 = 9 - 11x$$

$$10x^2 + 23x - 5 = 0$$

One root is $0.2 = \frac{1}{5}$.
The product of the roots is

$$\frac{-5}{10} = -\frac{1}{2}$$

So the other root is

$$\frac{-\frac{1}{2}}{\frac{1}{5}} = -\frac{5}{2}$$

Then

$$y = 3\left(-\frac{5}{2}\right) + 2$$

$$y = -\frac{11}{2}$$

FINAL ANSWER

$B = \left(-\frac{5}{2}, -\frac{11}{2}\right)$

PAST PAPER QUESTION

Q#: 28 **Marks: 6**

TOPIC: **Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

28 The curve **C** has equation $x^2 + y^2 = d - 11x$ where d is an integer.
The line **L** has equation $y = 3x + e$ where e is an integer.

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The coordinates of A are (0.2, 2.6)

Work out the coordinates of B
Show your working clearly.

(.......................... , ...)

(Total for Question 28 is 6 marks)

PAST PAPER SOLUTION

Q#: 28 Marks: 6

TOPIC: **Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The line is

$$y = 3x + e$$

Point $A = (0.2, 2.6)$ lies on the line.

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So the other root is

$$\frac{-\frac{1}{2}}{\frac{1}{5}} = -\frac{5}{2}$$

Then

$$y = 3\left(-\frac{5}{2}\right) + 2$$

$$y = -\frac{11}{2}$$

FINAL ANSWER
$B = \left(-\frac{5}{2}, -\frac{11}{2}\right)$